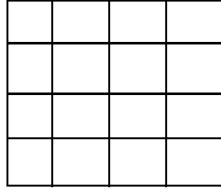


AMTI (NMTC) - 2004

GAUSS CONTEST - FINAL - PRIMARY LEVEL

1. Find all the three digit numbers formed by 3, 5 and 7 in which no digit is repeated. For example if you do the same for 1, 2, 3 we have 123,231,213 as some of the numbers that you can get. Add all of them and divide the sum by $3 + 5 + 7$. Call the number that you get as a. Now find all the three digit numbers that are formed by 2, 6 and 8, again without repetitions. Add all of them and divide the sum by $2 + 6 + 8$. Call the number you get as b. Compare a and b to find which is bigger.
2. Ram bought a notebook containing 98 pages, and numbered them from 1 to 196. Krishna tore 35 pages of Ram's notebook and added in 70 numbers he found on the pages. Could Krishna have got 2004 as the sum ?
3. There are 20 cities in a certain country. Every pair of cities is connected by an air route. How many air routes are there ?
4. Ram checks his purse and finds that he can buy an apple and three oranges or two apples for the money he has. I buy, from the same shop, two apples and two oranges for Rs. 16. How much my friend should pay when he buys three apples and two oranges from the same shop ?
5. Let $d(n)$ denote the number of divisors of a positive integer n . For example, $d(6) = 4$, $d(7) = 2$, $d(12) = 6$ as 1, 2, 3, 6 are the divisors of 6; 1, 7 are the divisors of 7; and 1, 2, 3, 4, 6, 12 are the divisors of 12. We note that $d(n) = 2$ if and only if n is a prime integer. Prepare a table which gives the values of $d(n)$ for $n = 1, 2, 3, \dots, 20$.
 - (a) Find $d(4)$, $d(49)$, $d(121)$ and $d(37 \times 37)$.
 - (b) Find n such that $1 \leq n \leq 100$ for which $d(n) = 3$.
 - (c) Use the table you have prepared above to find n between 2000 and 2009 such that $d(n)$ is an odd number.
 - (d) If $d(n)$ is a very big integer, then n is clearly a bigger integer. Looking in the opposite direction, if n is a very big integer can we say that $d(n)$ is at least half as big ?
 - (e) Can you find a big integer K such that for any integer n bigger than K we have $d(n) \geq 3$?
6. Some prime numbers are generated as follows. Start with a prime number. For example 3. Then consider $2 \times 3 + 1$. It is 7. It is a prime number. Again multiply by 2 and add 1 to get $2 \times 7 + 1$ to get 15. Now 15 is not a prime. So find the least prime dividing 15; which is the number 3. The sequence generated so far is 3, 7, 3. If we continue this process we will get the sequence 3, 7, 3, 7, 3, 7, 3, The process is given by
 - (a) Start with a prime number p_1 .
 - (b) Multiply p_1 by 2 and add 1 to get $2p_1 + 1$.
 - (c) If $2p_1 + 1$ is a prime write $2p_1 + 1 = p_2$.
 - (d) If $2p_1 + 1$ is not a prime call the smallest prime factor of $2p_1 + 1$ as p_2 .
 - (e) Multiply p_2 by 2 and add 1 to get $2p_2 + 1$.
 - (f) If $2p_2 + 1$ is a prime write $2p_2 + 1 = p_3$.
7. Consider the first five natural numbers 1, 2, 3, 4, 5. This set of five numbers is divided into two sets A and B where A contains two numbers and B contains the other three numbers. One example is $A = \{2, 4\}$ and $B = \{1, 3, 5\}$. How many such pairs A, B of sets are there ?

8. Draw a 4×4 square as shown. Fill the 16 squares with the letters a, b, c, d so that each letter appears exactly once in each row and also exactly once in each column. Give at least two different solutions.



9. One can easily see that if a perfect square n^2 is divisible by a prime p then it is also divisible by p^2 . For example any square integer that is divisible by 7 is also divisible by 49. Can a number written with 200 zeroes, 200 ones and 200 twos be a perfect square ?
10. The positive integers a and b satisfy $23a = 32b$. Can $a + b$ be a prime number ? Justify your answer.
11. Each square in a 2×2 table is coloured either black or white. How many different colourings of the table are there ?
12. Explain why an equilateral triangle (a triangle with equal sides) cannot be covered by two smaller equilateral triangles.
13. The side AC of a triangle is of length 2.7 cms., and the side AB has length 0.7 cms. If the length of the side BC is an integer, what is the length of BC ?
14. It is well known that the diagonals of a parallelogram bisect each other. In any triangle the line segment joining a vertex with the midpoint of the opposite side is called a median. If ABC is any triangle prove that the sum of the lengths of the three medians is not greater than the triangle's perimeter.
15. In the adjacent figure we have a star with five vertices A, B, C, D, E. Find the sum of the angles at the vertices A, B, C, D, E of the five pointed star. (You may use the fact that in any triangle the sum of the angles is 180°).

